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Idea Submission Template / Code. Create. Innovate.

Team Name	Team Raksha
Problem Statement No.	2
Problem Statement Title	Improving Safety at Large Public Events
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IDEA TITLE

Know the crowd. Catch the bottleneck. Stop the surge.

Cameras measure how dense each zone is, and how fast it moves.
It warns guards before a crush builds.

18

tracked at once

<1s

guard warned

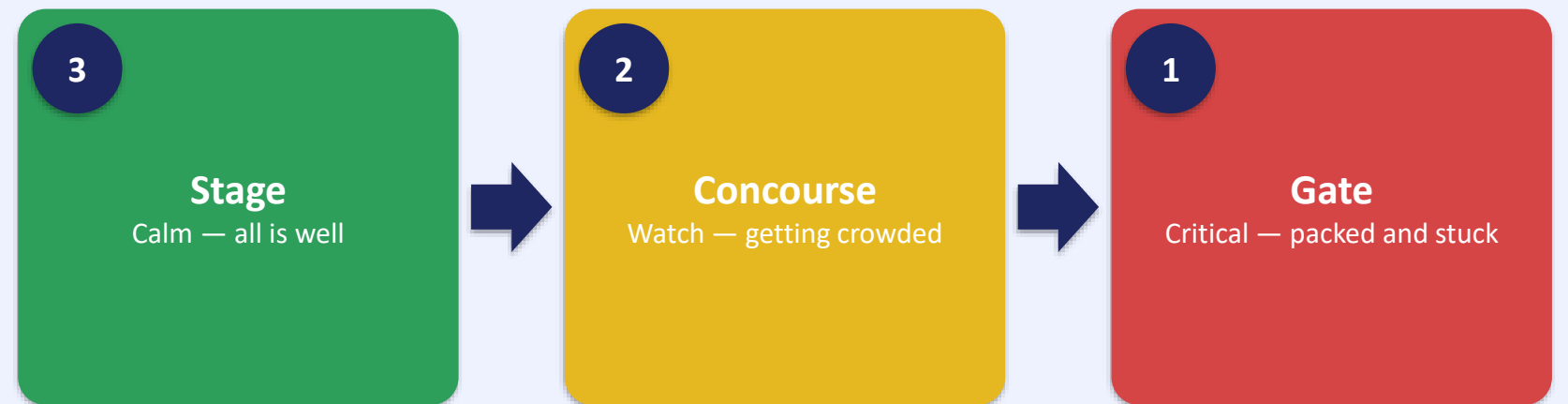
15

video checks every
second

Watch it work

- 1 Live feed: every person counted, zone density tracked.
- 2 A zone packs up and stops: bottleneck flagged.
- 3 Guard warned in seconds; wall shows time left to critical.

Venue map — every zone ranked by risk



Arrows follow the crowd. Risk ranks the other way: Gate first.

Guard screen: one map with every zone ranked by risk.

What the colours mean

Calm
All is well.

Watch
Getting crowded. Guards get
an early warning.

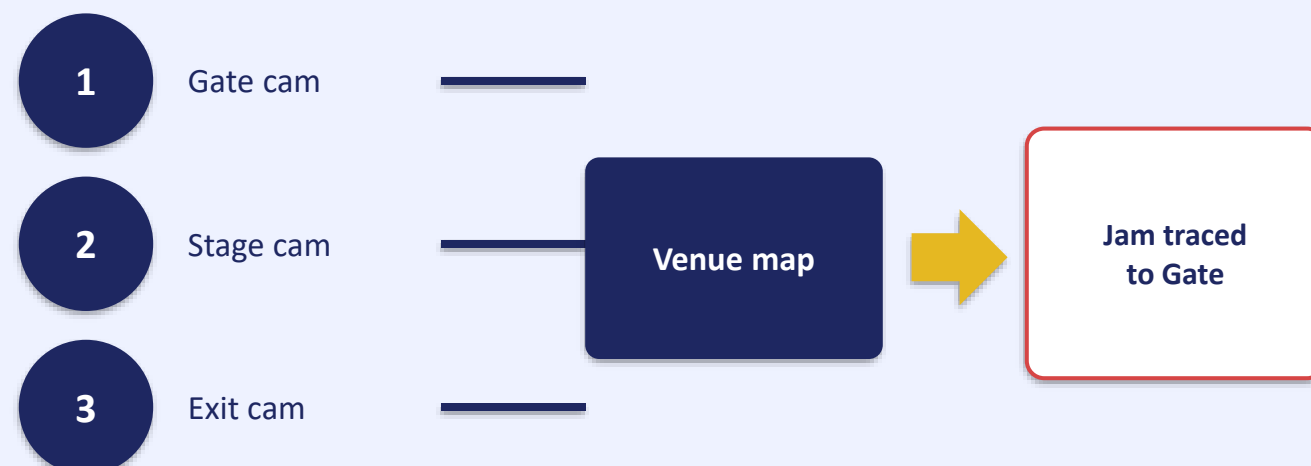
Critical
Crowd packed and stuck.
Clear the area now.

TECHNICAL APPROACH



In plain words: cameras share one venue map, the software flags where the crowd stops moving, and it warns guards before a crush builds.

One venue graph



Crowd metrics, not faces

Risk = density x stillness x counterflow. Stillness only escalates risk when the zone is also packed.

Tested on real video

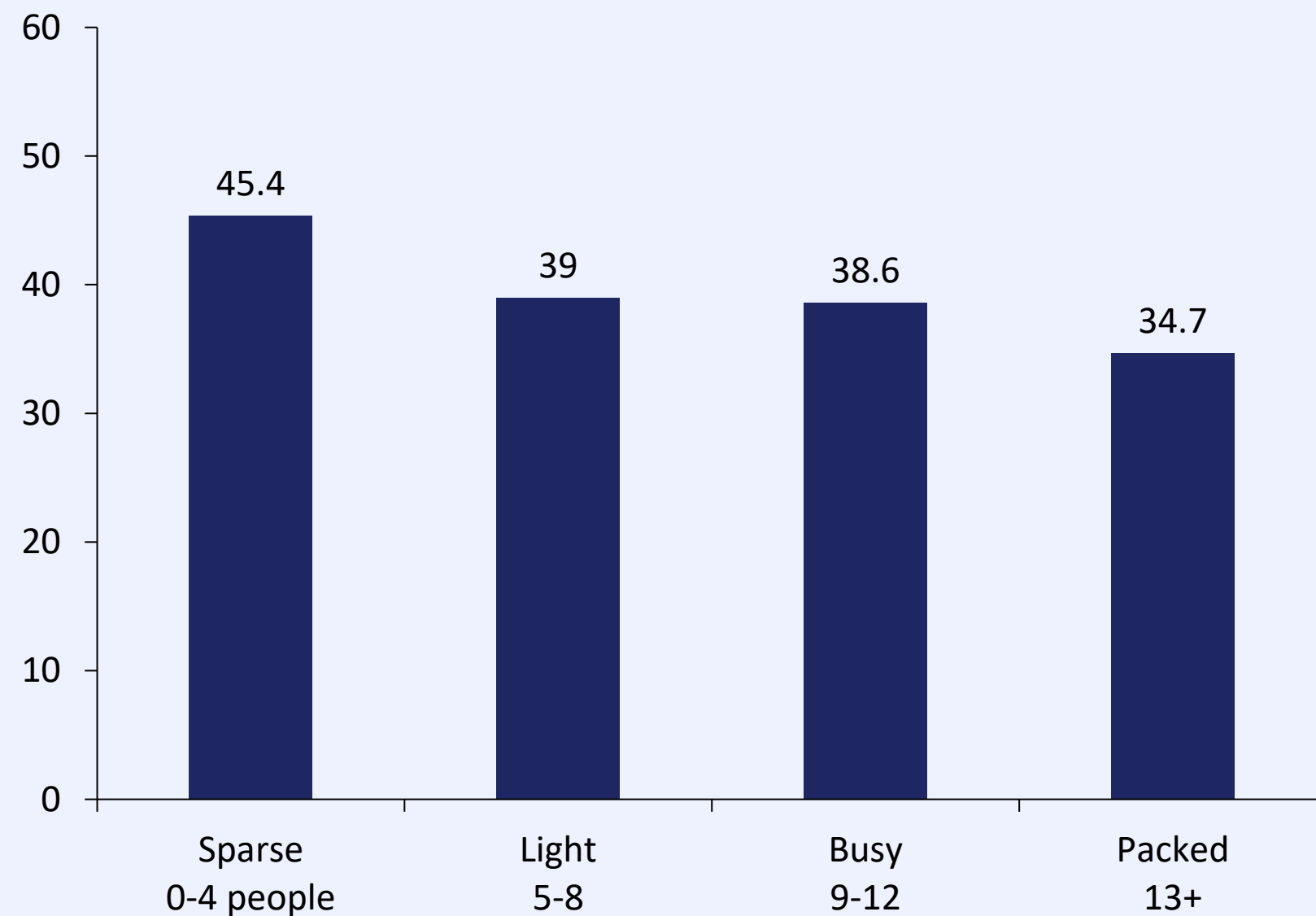
Followed 18 people at once across 2 minutes of busy station video.

Works with normal cameras

Cameras link into one venue graph, so a jam is traced back to where it started. High, wide angles measure density best.

FEASIBILITY AND VIABILITY

Checks per second as the crowd grows



Model and tracker time only, same laptop. FPS stays above 30 even when the frame fills with people.

Forecast

The trend is fitted live, so the wall shows seconds to critical before the jam locks.

Red alert

Fires when a zone packs tight and stops. The wall counts down the seconds to critical.

Built to scale out

One laptop runs a venue today; an edge box per camera scales to mass gatherings.

Tested on three real crowds

Station concourse

16 people peak

37 FPS

Busy crossing

7 people peak

42 FPS

Temple street

15 people peak

47 FPS

Same laptop, 500 frames per clip — model and tracker time only.

IMPACT AND BENEFITS

When the alarm goes up

Yellow watch

Guards keep eyes on the zone. Crowd flow is slowed.

Red alert

Guards get steps on their handsets; a human confirms before any PA or door release.

Handover

A sealed evidence pack goes to the police.

Pilot targets: tens of seconds forecast before critical, guard warned under 1 second, every step logged.

Before

A wall of screens, one tired guard, and risks get missed.

After

One screen ranks zones and traces jams to their origin: security moves from reactive to proactive.

Organizers

Spot a growing jam early, and tell guards which zone to clear.

Police

Every alert, zone and action exports as one evidence pack in a click.

Public

Faces stay blurred in saved video.

RESEARCH AND REFERENCES

The science behind it

Finding and following people

Finds every person and follows each one. Uses YOLOv8 and ByteTrack, the standard real-time pairing for crowd tracking.

Code: github.com/ultralytics/ultralytics, github.com/ifzhang/ByteTrack

Phase 2: face search

Not in the MVP. The pipeline is modular, so agencies can plug in their own face-search API later for post-incident cases (opt-in, B2G).

Code: github.com/deepinsight/insightface, github.com/facebookresearch/faiss

Predicting crushes

The AURORA warning model plus our live optical-flow stall meter spot danger early. A Delhi trial cut response time by 37 percent.

Ideas: AURORA warning model, optical-flow stall metrics, STAR-Crowd Delhi trial

Why India needs it

Packed crowds and blocked exits turned deadly at Hathras, Kumbh and railway stations.

Test video: [Wikimedia Commons](#) public footage (station, crossing, temple)

Privacy built in

Anonymized by default

Saved video is face-blurred. Live alerts show zones and counts, not names.

7-day routine wipe

Daily footage wipes after 7 days. Flagged incidents are exported first and kept for the police.

Tamper-evident log

Every action is hash-chained. Any edit leaves a trace.

No ID in core

Crowd mode runs with zero identity. Face search stays opt-in Phase 2 only.